



# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Revision Date: 03/21/2024

Version: 1.0

### SECTION 1: IDENTIFICATION

#### Product Identifier

Product Form: Mixture

Product Name: TM 2000 RB pH Plus

Product Code: AFCO 5654

#### Intended Use of the Product

Use of the Substance/Mixture: Water Treatment Compound. For professional use only.

#### Name, Address, and Telephone of the Responsible Party

##### Company

AFCO

550 Development Avenue

Chambersburg, PA 17201

T: 800-345-1329

[www.afcocare.com](http://www.afcocare.com)

#### Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

### SECTION 2: HAZARDS IDENTIFICATION

#### Classification of the Substance or Mixture

##### Classification (GHS-US)

Eye Irrit. 2A H319

Aquatic Acute 3 H402

Aquatic Chronic 3 H412

#### Label Elements

##### GHS-US Labeling

##### Hazard Pictograms (GHS-US):



##### Signal Word (GHS-US)

: Warning.

##### Hazard Statements (GHS-US)

: H319 - Causes serious eye irritation

H402 - Harmful to aquatic life

H412 - Harmful to aquatic life with long lasting effects

##### Precautionary Statements (GHS-US)

: P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.

P273 - Avoid release to the environment.

P280 - Wear protective glove, protective clothing, eye protection, face protection.

P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P337+P313 - If eye irritation persists: Get medical advice/attention.

P501 - Dispose of contents/container according to local, regional, national, territorial, provincial, and international regulations.

#### Other Hazards

Results of PBT and vPvB Assessment: This mixture does not contain any substances that are assessed to be a PBT or a vPvB.

Unknown Acute Toxicity (GHS-US): Not available.

### SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances: Not relevant (mixture).

#### Mixture

| Name  | Product identifier | % (w/w) | Classification (GHS-US) |
|-------|--------------------|---------|-------------------------|
| Water | (CAS No) 7732-18-5 | 65-80   | Not classified.         |

# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

|                                                               |                      |       |                                                                                                   |
|---------------------------------------------------------------|----------------------|-------|---------------------------------------------------------------------------------------------------|
| 2-phosphonobutane 1,2, 4 tricarboxylic acid, tetrasodium salt | (CAS No.) 66669-53-2 | 15-25 | Acute Tox. 4 (Oral) , H302<br>Eye Irrit. 2A, H319                                                 |
| Sodium carbonate                                              | (CAS No.) 497-19-8   | 1-5   | Eye Irrit. 2A, H319                                                                               |
| Sodium salt of hydroxyethylidene-1,1, diphosphonic acid       | (CAS No.) 29329-71-3 | 1-5   | Not classified.                                                                                   |
| Zinc sulfate heptahydrate                                     | (CAS No.) 7447-20-0  | <1.0  | Acute Tox. 4 (Oral), H302<br>Eye Dam. 1, H318<br>Aquatic Acute 1 ,H400<br>Aquatic Chronic 1, H410 |

The exact percentage of composition are withheld as a trade secret [29 CFR 1910.1200].

Full text of H-phrases: see section 16.

## SECTION 4: FIRST AID MEASURES

### Description of First Aid Measures

**General:** Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

**Inhalation:** When symptoms occur: go into open air and ventilate suspected area. Remove to fresh air and keep at rest in a position comfortable for breathing. If breathing is irregular or stopped, immediately call a POISON CENTER or doctor/physician.

**Skin Contact:** Remove contaminated clothing. Drench affected area with water for several minutes. If irritation persists, immediately call a POISON CENTER or doctor/physician. Wash contaminated clothing before reuse.

**Eye Contact:** Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If irritation occurs, immediately call a POISON CENTER or doctor/physician.

**Ingestion:** Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

### Most Important Symptoms and Effects Both Acute and Delayed

Symptoms and effects are not known to date.

### Indication of Any Immediate Medical Attention and Special Treatment Needed

None.

## SECTION 5: FIRE-FIGHTING MEASURES

### Extinguishing Media

**Suitable Extinguishing Media:** Water spray, fog, carbon dioxide, foam, dry chemical.

**Unsuitable Extinguishing Media:** Do not use a heavy water jet. Use of heavy stream of water may spread fire.

### Special Hazards Arising From the Substance or Mixture

**Fire Hazard:** Not flammable.

**Explosion Hazard:** Product is not explosive.

**Reactivity:** Thermal decomposition generates: Carbon oxides (CO, CO<sub>2</sub>), sulfur oxides (SO<sub>x</sub>), zinc oxides (ZnO<sub>x</sub>), phosphorus oxides (PO<sub>x</sub>). When heated to decomposition, emits toxic fumes.

### Advice for Firefighters

**Precautionary Measures Fire:** Exercise caution when fighting any chemical fire.

**Firefighting Instructions:** Use water spray or fog for cooling exposed containers.

**Protection During Firefighting:** Do not enter fire area without proper protective equipment, including respiratory protection.

**Hazardous Combustion Products:** Carbon oxides (CO, CO<sub>2</sub>), sulfur oxides (SO<sub>x</sub>), zinc oxides (ZnO<sub>x</sub>), phosphorus oxides (PO<sub>x</sub>).

**Other information:** Do not allow run-off from fire fighting to enter drains or water courses.

### Reference to Other Sections

Refer to section 9 for flammability properties.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

**General Measures:** Do not get on skin, eyes, or clothing. Do not allow product to spread into the environment.

### For Non-Emergency Personnel

**Protective Equipment:** Use appropriate personal protection equipment (PPE).

**Emergency Procedures:** Evacuate unnecessary personnel.

### For Emergency Personnel

**Protective Equipment:** Equip cleanup crew with proper protection.

**Emergency Procedures:** Ventilate area.

# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

### Environmental Precautions

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it.

### Methods and Material for Containment and Cleaning Up

**For Containment:** Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

**Methods for Cleaning Up:** Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill.

Collect spillage with sawdust, kieselgur (diatomite), sand, universal binder, Absorbents and binders.

### Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

**Additional Hazards When Processed:** When heated to decomposition, emits toxic fumes.

**Hygiene Measures:** Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do not eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling.

### Conditions for Safe Storage, Including Any Incompatibilities

**Technical Measures:** Comply with applicable regulations.

**Storage Conditions:** Store in a dry, cool and well-ventilated place. Keep container closed when not in use. Keep/Store away from extremely low temperatures, incompatible materials.

**Incompatible Materials:** Strong oxidizers.

**Recommended Storage Temperature:** 5-35°C/41-95°F

**Specific End Use(s)** Water treatment compound. For professional use only.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters:** None.

### Exposure Controls

**Appropriate Engineering Controls:** Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment:** Protective goggles. Gloves.



**Materials for Protective Clothing:** N/A.

**Hand Protection:** Wear chemically resistant protective gloves.

**Eye Protection:** Chemical goggles or face shield.

**Skin and Body Protection:** None usually needed.

**Respiratory Protection:** A respirator is not needed under normal and intended conditions of use. In case of inadequate ventilation wear respiratory protection.

**Other Information:** When using, do not eat, drink or smoke. Wash hands before breaks and at the end of workday.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on Basic Physical and Chemical Properties

|                                            |                                    |
|--------------------------------------------|------------------------------------|
| Physical State                             | : Liquid.                          |
| Appearance                                 | : Clear colorless to light yellow. |
| Odor                                       | : Characteristic.                  |
| Odor Threshold                             | : Not determined.                  |
| pH                                         | : 10.5                             |
| Relative Evaporation Rate (butylacetate=1) | : Not determined.                  |
| Melting Point                              | : Not determined.                  |
| Freezing Point                             | : Not determined.                  |
| Boiling Point                              | : Not determined.                  |
| Flash Point                                | : Not determined.                  |
| Auto-ignition Temperature                  | : Not determined.                  |

# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

|                                               |                   |
|-----------------------------------------------|-------------------|
| <b>Decomposition Temperature</b>              | : Not determined. |
| <b>Flammability (solid, gas)</b>              | : Not relevant.   |
| <b>Lower Flammable Limit</b>                  | : Not determined. |
| <b>Upper Flammable Limit</b>                  | : Not determined. |
| <b>Vapor Pressure</b>                         | : Not determined. |
| <b>Relative Vapor Density at 20°C</b>         | : Not determined. |
| <b>Specific Gravity</b>                       | : 1.29            |
| <b>Solubility</b>                             | : Not determined. |
| <b>Partition coefficient: n-octanol/water</b> | : Not determined. |
| <b>Viscosity</b>                              | : Not determined. |
| <b>Explosion Data</b>                         | : None.           |
| <b>Oxidizing Data</b>                         | : None.           |

## SECTION 10: STABILITY AND REACTIVITY

**Reactivity:** Thermal decomposition generates: Carbon oxides (CO, CO<sub>2</sub>), sulfur oxides (SO<sub>x</sub>), zinc oxides (ZnO<sub>x</sub>), phosphorus oxides (PO<sub>x</sub>).

**Chemical Stability:** Stable under normal conditions.

**Possibility of Hazardous Reactions:** Hazardous polymerization will not occur.

**Conditions to Avoid:** Extremely low temperatures. Incompatible materials.

**Incompatible Materials:** Strong oxidizers.

**Hazardous Decomposition Products:** Carbon oxides (CO, CO<sub>2</sub>), sulfur oxides (SO<sub>x</sub>), zinc oxides (ZnO<sub>x</sub>), phosphorus oxides (PO<sub>x</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects - Product

**Acute Toxicity:** Not classified.

**LD50 and LC50 Data:** Not determined.

**Skin Corrosion/Irritation:** Not classified.

**Serious Eye Damage/Irritation:** Causes serious eye irritation.

**Respiratory or Skin Sensitization:** Not classified.

**Germ Cell Mutagenicity:** Not classified.

**Teratogenicity:** Not classified.

**Carcinogenicity:** Not classified.

**Specific Target Organ Toxicity (Repeated Exposure):** Not classified.

**Reproductive Toxicity:** Not classified.

**Specific Target Organ Toxicity (Single Exposure):** Not classified.

**Aspiration Hazard:** Not classified.

**Symptoms/Injuries After Inhalation:** None under normal and intended conditions of use.

**Symptoms/Injuries After Skin Contact:** May cause mild skin irritation.

**Symptoms/Injuries After Eye Contact:** Serious eye irritation including redness and watering of eyes.

**Symptoms/Injuries After Ingestion:** Ingestion is likely to be harmful or have adverse effects.

### Information on Toxicological Effects - Ingredient(s)

**LD50 and LC50 Data:**

|                                                                             |              |
|-----------------------------------------------------------------------------|--------------|
| <b>Water (7732-18-5)</b>                                                    |              |
| LD50 Oral Rat                                                               | >90000 mg/kg |
| <b>Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)</b> |              |
| LD50 Oral Rat                                                               | 940 mg/kg    |
| LD50 Dermal Rabbit                                                          | >1650 mg/kg  |
| <b>Zinc sulfate heptahydrate (7446-20-0)</b>                                |              |
| LD50 Oral Rat                                                               | 2150 mg/kg   |
| LD50 Intraperitoneal Rat                                                    | 200 mg/kg    |
| <b>Sodium carbonate (497-19-8)</b>                                          |              |
| LD50 Oral Rat                                                               | 2800 mg/kg   |
| LD50 Dermal Rabbit                                                          | >2000 mg/kg  |

# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

### SECTION 12: ECOLOGICAL INFORMATION

#### Toxicity

**PRODUCT AS SOLD** - Not determined.

#### PRODUCT INGREDIENTS

|                                                                             |                                                                                            |
|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)</b> |                                                                                            |
| LC50 Fish 1                                                                 | 368 mg/l (Exposure time: 96h - Species: Rainbow trout)                                     |
| EC50 Daphnia 1                                                              | 527 mg/l (Exposure time: 48h - Species: Daphnia magna)                                     |
| EC50 Algae                                                                  | 9.16 mg/l (Exposure time: 96h - Species: Selenastrum capricornutum)                        |
| <b>Zinc sulfate heptahydrate (7446-20-0)</b>                                |                                                                                            |
| LC50 Fish 1                                                                 | 1-10 mg/l (Exposure time: 96h - Species: other fish)                                       |
| Aquatic invertebrate                                                        | Not determined                                                                             |
| <b>Sodium carbonate (497-19-8)</b>                                          |                                                                                            |
| LC50 Fish 1                                                                 | 300 mg/l (Exposure time: 96h - Species: Lepomis macrochirus) (Bluegill sunfish) (static)   |
| EC50 Daphnia 1                                                              | 200-227 mg/l (Exposure time: 48h - Species: Ceriodaphnia dubia) (water flea) (semi-static) |

#### Persistence and Degradability

|                                       |                  |
|---------------------------------------|------------------|
| <b>TM 2000 RB pH Plus (AFCO 5654)</b> |                  |
| Persistence and Degradability         | Not established. |

#### PRODUCT INGREDIENTS

|                                                                             |  |
|-----------------------------------------------------------------------------|--|
| <b>Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)</b> |  |
| Closed bottle BOD/COD 5% - degrades after acclimatization.                  |  |
| <b>Sodium carbonate (497-19-8)</b>                                          |  |
| Abiotic degradation (Photodegradation) Hydrolyzes in water.                 |  |
| Biodegradation – Not applicable. Inorganic substance.                       |  |

#### Bioaccumulative Potential

|                                       |                  |
|---------------------------------------|------------------|
| <b>TM 2000 RB pH Plus (AFCO 5654)</b> |                  |
| Bioaccumulative Potential             | Not established. |

#### PRODUCT INGREDIENTS

|                                                                             |  |
|-----------------------------------------------------------------------------|--|
| <b>Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)</b> |  |
| Not expected to bioaccumulate.                                              |  |
| <b>Sodium carbonate (497-19-8)</b>                                          |  |
| Not applicable. Inorganic substance.                                        |  |

#### Mobility in Soil

|                                  |                  |
|----------------------------------|------------------|
| <b>TM 2000 RB pH Plus (####)</b> |                  |
| Mobility in Soil                 | Not established. |

#### PRODUCT INGREDIENTS

|                                    |  |
|------------------------------------|--|
| <b>Sodium carbonate (497-19-8)</b> |  |
| <b>Adsorption potential (Koc):</b> |  |
| Air- Not applicable                |  |
| Solubility - water                 |  |
| Mobility - water                   |  |
| Soil/sediment - Not significant    |  |

#### Other Adverse Effects

**Other Information:** Avoid release to the environment.

### SECTION 13: DISPOSAL CONSIDERATIONS

**Waste Disposal Recommendations:** Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

### SECTION 14: TRANSPORT INFORMATION

|                              |   |                              |
|------------------------------|---|------------------------------|
| 14.1 In Accordance with DOT  | : | Not regulated for transport. |
| 14.2 In Accordance with IMDG | : | Not regulated for transport. |
| 14.3 In Accordance with IATA | : | Not regulated for transport. |
| 14.4 In Accordance with TDG  | : | Not regulated for transport. |

### SECTION 15: REGULATORY INFORMATION

#### US Federal Regulations

|                                                                                               |                                  |
|-----------------------------------------------------------------------------------------------|----------------------------------|
| TM 2000 RB pH Plus (AFCO 5654)                                                                |                                  |
| All components are listed on the United States TSCA (Toxic Substances Control Act) inventory. |                                  |
| SARA Section 311/312 Hazard Class                                                             | Immediate (acute) health hazard. |

**US State Regulations:** No components listed.

#### Canadian Regulations

|                                                                           |  |
|---------------------------------------------------------------------------|--|
| Water (7732-18-5)                                                         |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.          |  |
| Sodium carbonate (497-19-8)                                               |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.          |  |
| Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)      |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.          |  |
| 2-phosphonobutane 1,2, 4 tricarboxylic acid, tetrasodium salt (6666-53-2) |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.          |  |
| Zinc sulfate heptahydrate (7446-20-0)                                     |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.          |  |

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

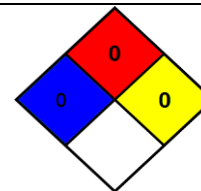
### SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

|                   |   |                                                                                                                                     |
|-------------------|---|-------------------------------------------------------------------------------------------------------------------------------------|
| Revision date     | : | 03/21/2024                                                                                                                          |
| Other Information | : | This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200. |

#### GHS Full Text Phrases:

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| Acute Tox. 4 (Oral) | Acute toxicity (oral) Category 3.                                 |
| Aquatic Acute 1     | Hazardous to the aquatic environment - Acute Hazard Category 1.   |
| Aquatic Acute 3     | Hazardous to the aquatic environment - Acute Hazard Category 3.   |
| Aquatic Chronic 1   | Hazardous to the aquatic environment - Chronic Hazard Category 1. |
| Aquatic Chronic 3   | Hazardous to the aquatic environment - Chronic Hazard Category 2. |
| Eye Dam. 1          | Serious eye damage/eye irritation Category 1.                     |
| Eye Irrit. 2A       | Serious eye damage/eye irritation Category 2A.                    |
| H302                | Harmful if swallowed.                                             |
| H318                | Causes serious eye damage.                                        |
| H319                | Causes serious eye irritation.                                    |
| H400                | Very toxic to aquatic life.                                       |
| H402                | Harmful to aquatic life.                                          |
| H410                | Very toxic to aquatic life with long lasting effects.             |
| H412                | Harmful to aquatic life with long lasting effects.                |

|                    |   |                                                                                                         |
|--------------------|---|---------------------------------------------------------------------------------------------------------|
| NFPA Health Hazard | : | 0 - Exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials. |
| NFPA Fire Hazard   | : | 0 - Materials that will not burn.                                                                       |
| NFPA Reactivity    | : | 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.              |



# TM 2000 RB pH Plus (AFCO 5654)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

---

### HMIS III Rating

- Health : 0 - Minimal Hazard - No significant risk to health.
- Flammability : 0 - Minimal Hazard.
- Physical : 0 - Minimal Hazard.

### Party Responsible for the Preparation of This Document

AFCO  
550 Development Avenue  
Chambersburg, PA 17201  
T: 800-345-1329

*This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.*

North America GHS SDS 2015 (U.S., Can., Mex.)